Problem Set 5D

Vector equations of planes

Do these after the slides for Section 5D. Textbook Exercise 5D. Answers are on the last page. Show full working in your exercise book.

Skills

1 Find the Cartesian equation of the plane through $(2, -1, 3)$ with normal $4\mathbf{i} + \mathbf{j} - 2\mathbf{k}$.

2 For the plane $3x - 2y + z = 6$:

(a) write the equation in the form $\mathbf{r} \cdot \mathbf{n} = k$

(b) find a unit normal

(c) find where the plane cuts the x-axis

3 Find the Cartesian equation of the plane through $A(1, 1, 0)$, $B(2, 0, 3)$ and $C(0, 2, 1)$.

4 Find the equation of the plane through $(1, 2, 3)$ parallel to the plane $2x - y + 4z = 7$.

5 Find the equation of the plane through $(3, 0, -1)$ perpendicular to the line $\mathbf{r} = \mathbf{i} + \mathbf{j} + \mathbf{k} + t(2\mathbf{i} - 3\mathbf{j} + \mathbf{k})$.

Exam style

6 The lines $\mathbf{r} = \mathbf{i} + \mathbf{j} + \lambda(\mathbf{i} + 2\mathbf{j} + \mathbf{k})$ and $\mathbf{r} = -2\mathbf{i} + 5\mathbf{j} - 5\mathbf{k} + \mu(2\mathbf{i} - \mathbf{j} + 3\mathbf{k})$ intersect. Find the Cartesian equation of the plane containing both lines.

7 Find the Cartesian equation of the plane containing the point $A(1, 0, 2)$ and the line $\mathbf{r} = 2\mathbf{i} + \mathbf{j} + t(\mathbf{i} + \mathbf{j} - \mathbf{k})$.

8 The point $(3, k, 2)$ lies on the plane through $(0, 0, 1)$, $(1, 1, 0)$ and $(2, 0, 3)$. Find k.

Problem Set 5D: answers

1 $4x + y - 2z = 1$

2 (a) $\mathbf{r} \cdot (3\mathbf{i} - 2\mathbf{j} + \mathbf{k}) = 6$ (b) $\frac{1}{\sqrt{14}}(3\mathbf{i} - 2\mathbf{j} + \mathbf{k})$ (c) $(2, 0, 0)$

3 $\mathbf{n} = -4\mathbf{i} - 4\mathbf{j}$, so $x + y = 2$

4 $2x - y + 4z = 12$

5 $2x - 3y + z = 5$

6 $\mathbf{n} = 7\mathbf{i} - \mathbf{j} - 5\mathbf{k}$: $7x - y - 5z = 6$

7 $\mathbf{n} = \mathbf{i} - \mathbf{j}$: $x - y = 1$

8 Plane $x - 2y - z = -1$, so $k = 1$